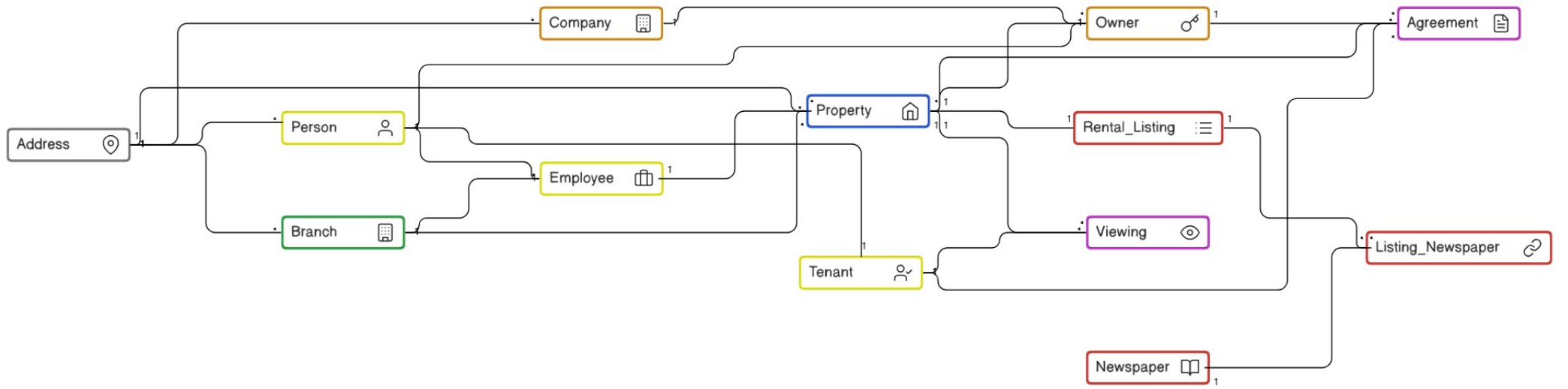




Branch (1) → (N) Property

One branch manages many properties.

Employee (0..1) → (1) Branch

An employee may manage one branch.

Person (1) → (0..1) Tenant

A person may be a tenant.

Person (1) → (0..N) Owner

A person may act as an owner.

Property (1) → (0..1) Rental_Listing

A property may have at most one listing.

Tenant (1) → (N) Viewing

A tenant can view many properties.

Tenant (1) → (N) Agreement

A tenant can have multiple agreements.

Property (M) → (N) Owner

A property can have multiple owners.

Newspaper (1) → (N) Listing_Newspaper

A newspaper can contain multiple listings.

Address (1) → (N) Person

An address can belong to many persons.

Address (1) → (N) Branch

An address can belong to many branches.

Branch (1) → (N) Employee

A branch employs multiple employees.

Employee (1) → (N) Property

An employee may manage multiple properties.

Person (1) → (0..1) Employee

A person may be an employee.

Company (1) → (0..N) Owner

A company may act as an owner.

Property (1) → (N) Viewing

A property can be viewed many times.

Property (1) → (N) Agreement

A property can have multiple agreements.

Owner (1) → (N) Agreement

An owner can sign multiple agreements.

Rental_Listing (1) → (N) Listing_Newspaper

A listing can be advertised multiple times.

Listing_Newspaper → Listing & Newspaper

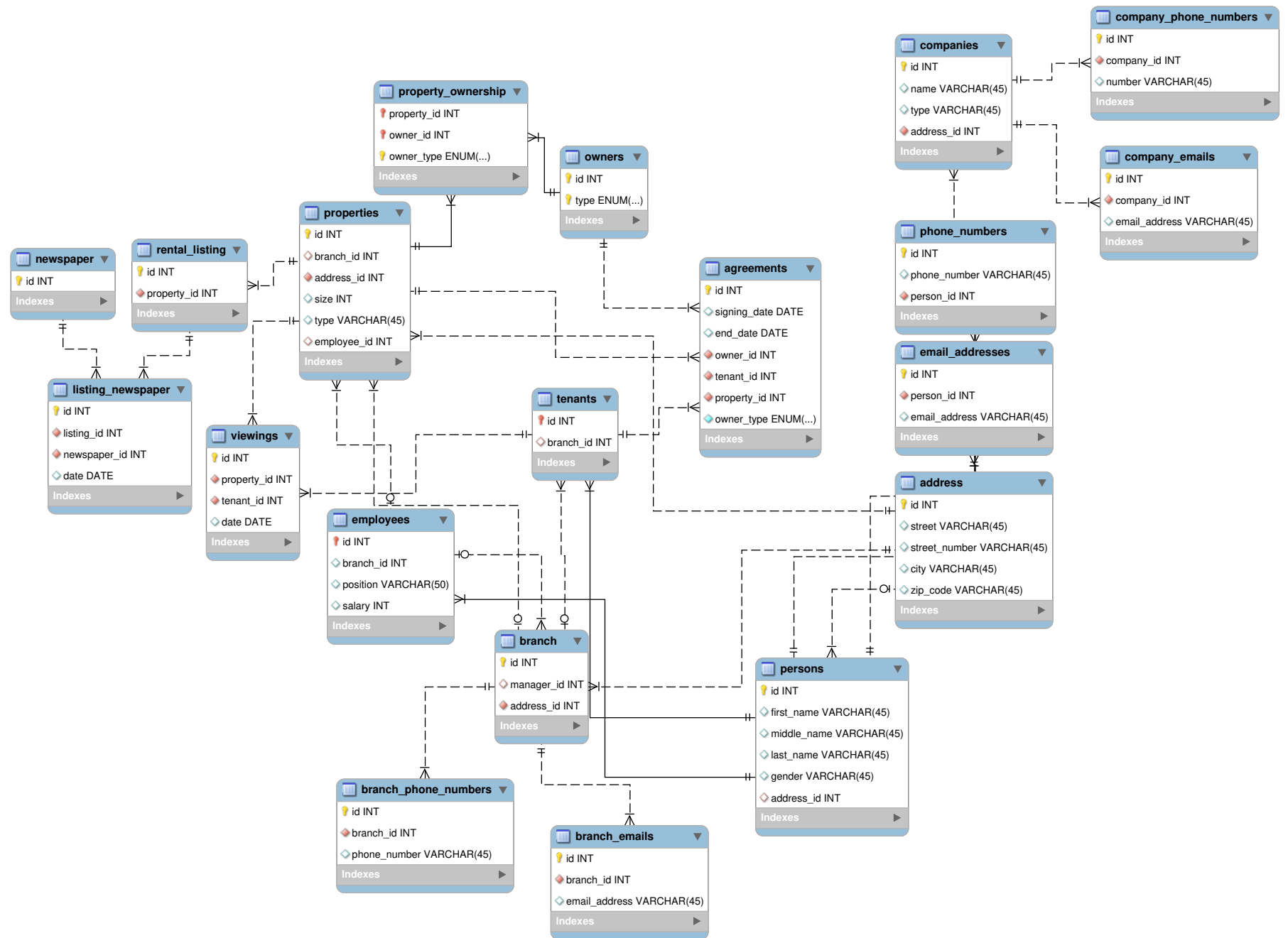
Represents listing appearance with date.

Address (1) → (N) Property

An address can belong to many properties.

Address (1) → (N) Company

An address can belong to many companies.



The application is a console-based property rental management system that operates through menu-driven input. The user interacts with the system by selecting options and entering required IDs and data step by step.

To use the system, the user first creates base entities: insert addresses, then create persons and companies linked to those addresses. Next, branches are created using an address, and employees are added (linked to persons) and optionally assigned to manage a branch. After that, properties are created and assigned to a branch, an address, and optionally an employee.

Once the base data is set up, the rental workflow can be used. The user can create rental listings for properties, record viewings by selecting a property and tenant, and create agreements by specifying owner, tenant, property, and dates. Owners (person or company) must be linked to properties before creating agreements.

Additional features allow linking listings to newspapers to track advertisements, and updating or querying existing records using IDs. The system relies on correct ID input and follows a logical order of entity creation to maintain valid relationships.

How to run the application

Start XAMPP and run Apache and MySQL, then open phpMyAdmin and import the provided db.sql file to create the database. After the database is set up, run the application by executing the main class (e.g., Main.java) in your IDE. The user can then navigate the system through the console by entering the number corresponding to each menu option.

```
--- MAIN MENU ---
```

1. Insert
2. Show
3. Update
4. Delete
0. Exit

Choose option:

Displays the primary navigation options of the application, allowing the user to choose between inserting, viewing, updating, or deleting data, or exiting the system.

```
--- UPDATE MENU ---
```

1. Person
2. Property
3. Agreement
4. Branch
5. Contact/Address
0. Go Back

Choose:

Allows the user to modify existing records such as persons, properties, agreements, branches, and contact/address information.

```
--- SHOW MENU ---
```

1. Available Properties
2. Rented Properties
3. Search For Available Property
4. Advertised Properties
5. Tenants With Multiple Viewings
0. Go Back

Choose: |

Displays query options for retrieving data, including available properties, rented properties, filtered property searches, advertised listings, and tenants with multiple viewings.

```
--- INSERT MENU ---
```

1. Person
2. Property
3. Rental listing
4. Agreement
5. Branch
6. Contact
7. Ownership
8. Viewing
0. Go Back

Choose:

Provides options for adding new records into the system, including persons, properties, rental listings, agreements, branches, contacts, ownerships, and viewings.

```
--- INSERT MENU ---
```

1. Person
2. Property
3. Rental listing
4. Agreement
5. Branch
6. Contact
7. Ownership
8. Viewing
0. Go Back

```
Choose: 1
```

1. Tenant
2. Owner
3. Employee
0. Go Back

```
Choose: 1
```

```
Enter person id or 0 to create new person
```

Shows the process of inserting a new person into the system, where the user selects the role (tenant, owner, or employee) and either enters an existing person ID or chooses to create a new person record.

